



RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 4 (Summary and Tutorial)

Topic: Sequence and Series, AP and GP

Summary for Summation

The Σ (Sigma) Notation

The series $u_1 + u_2 + \dots + u_n$ can be denoted by $\sum_{r=1}^n u_r$, i.e. $\sum_{r=1}^n u_r = u_1 + u_2 + \dots + u_n$.

In general, $\sum_{r=k}^m u_r$ ← Ending value of r
← General Term
← Starting value of r
 $= u_k + u_{k+1} + \dots + u_{m-1} + u_m$, where $k, m \in \mathbb{Z}_0^+$ and $k \leq m$.

e.g. $\sum_{r=3}^{n+1} (3r+1) = [3(3)+1] + [3(4)+1] + \dots + [3(n+1)+1] = 10 + 13 + \dots + (3n+4)$

e.g. $3^3 + 7^3 + 11^3 + \dots + 187^3 = \sum_{r=0}^{46} (3+4r)^3$ or $\sum_{r=1}^{47} (3+4(r-1))^3$

Properties of the Σ (Sigma) Notation

If u_r and v_r are expressions in terms of r , a and b are constants and k and m are non-negative integers such that $k \leq m$, then we have the following results:

(1) $\sum_{r=k}^m u_r$ has $(m-k+1)$ terms. (3) $\sum_{r=k}^m (au_r) = a \sum_{r=k}^m u_r$.	(2) For $2 \leq k \leq m$, $\sum_{r=k}^m u_r = \left(\sum_{r=1}^m u_r \right) - \left(\sum_{r=1}^{k-1} u_r \right)$. (4) $\sum_{r=k}^m (au_r \pm bv_r) = a \left(\sum_{r=k}^m u_r \right) \pm b \left(\sum_{r=k}^m v_r \right)$.
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Note that $\sum_{r=k}^m (u_r v_r) \neq \left(\sum_{r=k}^m u_r \right) \left(\sum_{r=k}^m v_r \right)$.

e.g.

(1): $\sum_{r=3}^{n+1} (3r+1)$ has $(n+1)-3+1 = n-1$ terms, if each $(3r+1)$ is considered as a term.

(2): $\sum_{r=3}^{n+1} (3r+1) = \sum_{r=1}^{n+1} (3r+1) - \sum_{r=1}^2 (3r+1)$.

(3) and (4): $\sum_{r=3}^{n+1} (3r+1) = 3 \sum_{r=3}^{n+1} r + \sum_{r=3}^{n+1} 1$.

Summary for Arithmetic and Geometric Progressions

Arithmetic and Geometric Progressions

Let a be the first term of the sequence, and d and r be the common difference and common ratio respectively.

	AP (d, n, a)	GP (r, a, n)
n^{th} term, u_n	$u_n = a + (n-1)d$	$u_n = ar^{n-1}$
	$u_n = S_n - S_{n-1}, n \geq 2$ $u_1 = S_1$	
Sum of first n terms, S_n	$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(u_1 + u_n)$	$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1}, r \neq 1$
To show sequence is a AP/GP	$u_{n+1} - u_n = \text{constant} (= d)$	$\frac{u_{n+1}}{u_n} = \text{constant} (= r)$
Sum to infinity, S_∞	-	$S_\infty = \frac{a}{1-r}, r < 1$

Special Sums

Let a, d and r be constants and k and n be positive integers such that $k \leq n$, then we have the following results:

$$(1) \sum_{k=1}^n [a + (k-1)d] = \frac{n}{2} [1^{\text{st}} \text{ term} + \text{last term}] = \frac{n}{2} [2a + (n-1)d].$$

LHS is the sum of 1st n terms of an **arithmetic** series with a as its first term and d as its common difference.

$$(2) \sum_{k=1}^n ar^{k-1} = \frac{a(r^n - 1)}{r - 1} = \frac{a(1 - r^n)}{1 - r}, r \neq 1.$$

LHS is the sum of 1st n terms of a **geometric** series with a as its first term and r as its common ratio.

Revision Tutorial Questions**Source of Question: VJC/Promo/2024/Q4**

- 1 (a) The sum, S_n , of the first n terms of a sequence $u_1, u_2, u_3, \dots$ is given by

$$S_n = 75 - \frac{3^{n+1}}{5^{n-2}}.$$

Show that $u_n = k\left(\frac{3}{5}\right)^{n-1}$, where k is a constant to be determined. Find S_∞ . [4]

- (b) Another sequence of numbers $v_1, v_2, v_3, \dots$ is such that $\sum_{i=1}^n v_i = \frac{2n}{2n+1}$.

Find the exact value of $\sum_{i=1}^{\infty} v_i$. [3]

Source of Question: RVHS/Promo/2024/Q4

- 2 It is given that $\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$.

- (a) Find $\sum_{r=n}^{2n} (r+1)^3$ where $n > 1$. Express your answer in the form

$$\frac{1}{a}(n+1)^2[a(2n+1)^2 - n^2] \text{ where } a \text{ is a constant to be determined.} [4]$$

- (b) Find $1^3 + 3^3 + 5^3 + \dots + (2k-3)^3 + (2k-1)^3$ where k is a positive integer. [2]

Source of Question: NYJC/Promo/2024/Q4

- 3 During a clinical trial, the concentration U , of a specific blood agent is measured at one-hour intervals following the initial administration of the trial drug to a patient.

The following readings were obtained

$$U_3 = 88, U_4 = 76 \text{ and } U_5 = 70,$$

where U_t denotes the reading t hours after the drug was first administered.

It is believed that U satisfies the relationship

$$U_{t+1} = p + qU_t, t \geq 0,$$

where p and q are constants.

- (a) Find the values of p and q . [2]
(b) Determine the initial concentration of the blood agent when the drug was initially administered. [3]
(c) Describe how the concentration behaves over a long period of time. [1]

Source of Question: DHS/JC1 Mid-Year CT/2018/01/Q4

4 The sum of the first n terms of a series is given by $20 - 5\left(\frac{3^n}{4^{n-1}}\right)$.

(i) Show that the series is a geometric series. [3]

(ii) Explain why this series converges and find its sum to infinity. [2]

A new geometric series is formed by using the even-numbered terms (i.e. $u_2, u_4, u_6, \dots$) of the above series. Find the sum of the first 20 terms of this new series. [2]

Source of Question: TJC/Prelim/2018/02/Q2

5 A convergent geometric progression has first term a and common ratio r , where $a > 0$ and $r > 0$. An arithmetic progression has positive first term b and non-zero common difference d . It is given that the 6th, 9th and 16th terms of the geometric progression are equal to the 7th, 10th and 13th terms of the arithmetic progression respectively.

(a) Find d in terms of b and r . Hence show that the arithmetic progression is a decreasing sequence. [4]

(b) (i) Show that r satisfies the equation

$$r^{10} - 2r^3 + 1 = 0.$$

Hence find the value of r , giving your answer correct to 3 significant figures. [3]

(ii) Using the value of r found in (i), find the least value of n for which the sum to infinity of the geometric series exceeds the sum of the first n terms by less than $0.006a$. [3]

Source of Question: SRJC/Prelim/2018/01/Q6

6 (a) (i) The twelfth, eighth and fifth terms of an arithmetic progression are three consecutive terms of a converging geometric progression of positive terms with common ratio r . Find the value of r . [3]

(ii) For this part, take the value of r to be $\frac{3}{4}$. If the difference between the sum of the first n terms of the geometric progression and its sum to infinity is less than 0.3% of the sum to infinity, find the least value of n . [3]

(b) A convergent geometric sequence of positive terms, G has a non-zero first term a and common ratio r .

(i) The sum of the first n odd-numbered terms of G is equal to the sum of all terms after the $(2n-1)$ th term of G . Show that $2r^{2n} + r^{2n-1} - 1 = 0$. [2]

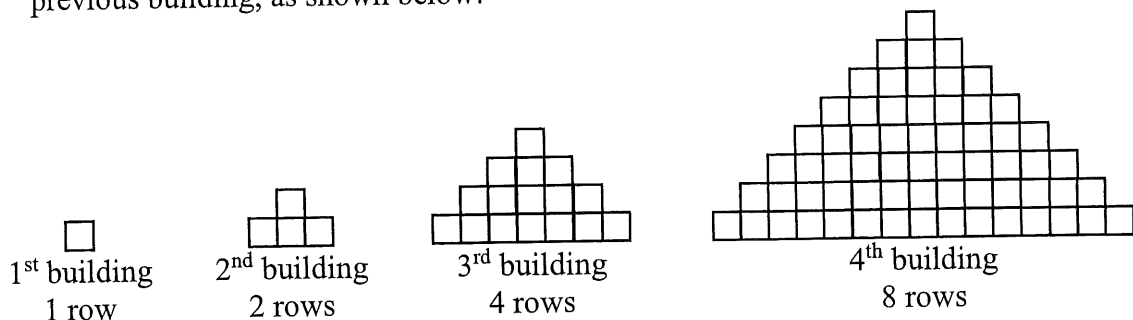
- (ii) In another sequence H , each term is the reciprocal of the corresponding term of G . If the n th term of G and H is denoted by u_n and v_n respectively, show that a new sequence whose n th term is $\ln\left(\frac{u_n}{v_n}\right)$, is an arithmetic progression. [2]

Source of Question: MJC/Prelim/2018/01/Q10 (modified)

- 7 (a) The sum of the first n terms of a sequence $\{u_n\}$ is given by $S_n = kn^2 - 3n$, where k is a non-zero real constant.
- (i) Prove that the sequence $\{u_n\}$ is an arithmetic sequence. [3]
- (ii) Given that u_2 , u_3 and u_6 are consecutive terms in a geometric sequence, find the value of k . [3]
- (b) A zoology student observes jaguars preying on white-tailed deer in the wild. He observes that when a jaguar spots its prey from a distance of d m away, it starts its chase. At the same time, the white-tailed deer senses danger and starts escaping.
- He models the predator-prey movements as follows:
- The jaguar starts its chase with a leap distance of 6 m. Subsequently, each leap covers a distance of 0.1 m less than its preceding leap.
- The white-tailed deer starts its escape with a leap distance of 9 m. Subsequently, each leap covers a distance of 5% less than its preceding leap.
- (i) Find the total distance travelled by a white-tailed deer after n leaps. Deduce the theoretical maximum total distance travelled by a white-tailed deer if it does not stop running. [3]
- (ii) Assume that both predator and prey complete the same number of leaps in the same duration of time. Given that $d = 11$ m, find the least value of n for a jaguar to catch a white-tailed deer within n leaps. [3]

Source of Question: HCI/JC2 CT1/2018/01/Q7

- 8 (a) A geometric progression has first term a and common ratio -3 . The first two terms of the geometric progression are the first and third terms of an arithmetic progression respectively. Find the sum of the first n even-numbered terms of the arithmetic progression in terms of a and n . [4]
- (b) A child constructs 'buildings' using wooden blocks so that each row has 2 more blocks than the row above, and each building has twice the number of rows as the previous building, as shown below:



Find expressions, in terms of r , for

- (i) the number of rows in the r^{th} building, [1]
 (ii) the number of blocks in the r^{th} building, [2]
 (iii) the total number of blocks in the first r buildings. [2]

Source of Question: TPJC/Prelim/2018/01/Q10

- 9 On 1 January 2018, Amy deposits \$200 into a bank account. On the first day of each subsequent month, she deposits \$20 more than in the previous month. Thus on 1 February, she deposits \$220 into the account and on 1 March, she deposits \$240 into the account, and so on. The account pays no interest.

- (i) On what date will the value of Amy's account first exceed \$6000? [5]

On 1 January 2018, Benjamin deposits \$200 into a savings account which pays compound interest at a rate of 0.4% per month on the last day of each month. He puts a further \$200 into the account on the first day of each subsequent month.

- (ii) Show that the value of Benjamin's savings account, in dollars, on the last day of the n^{th} month is given by

$$50200(1.004^n - 1). \quad [2]$$

- (iii) After how many complete months will the value of Benjamin's savings account first exceed \$6000? [3]

- (iv) Benjamin would like the value of his savings account to be \$6000 on 2 December 2019. What interest rate per month, applied from January 2018, would achieve this? [3]

Source of Question: MI/Prelim/2018/01/Q9

- 10 A toy all-terrain vehicle, Terra is designed to handle multiple terrains. In a test run, the vehicle moves on a path that consists of different sections of increasing difficulty. Terra completes the first section, second section, third section and so on until all sections have been completed. The total time (in seconds) taken by Terra to complete n sections is given by $\frac{3}{2}n^2 + \frac{13}{2}n$.

- (i) Find the time taken by Terra to complete the n th section. [2]

Terra was made to travel up a man-made hill of height 35 m, from the foot of the hill to the peak. Terra is designed to cover a vertical distance of 5 m in the first ten minutes. For each subsequent ten-minute period, Terra covers 80% of the vertical distance covered in the previous ten-minute period.

- (ii) The vertical distances covered by Terra in the ten-minute periods form a geometric sequence. Explain if this sequence is convergent. [1]
 (iii) Explain if Terra is able to reach the peak of the hill. [2]

After the completion of the fourth ten-minute period, Terra's vertical distance covered is changed such that in each subsequent ten-minute period, Terra covers 90% of the vertical distance covered in the previous ten-minute period.

- (iv) Find the minimum number of ten-minute periods required for Terra to reach the peak of the hill. [3]

After installing the brake system on Terra, the design team wants to test its effectiveness. In this test, Terra moves with increasing speed from a stationary position for 4 seconds before the brake system is engaged. Terra then has to stop moving in 2 seconds. The test is done many times and is repeated immediately after the previous one.

Terra's motion is modelled by the function given by

$$f(t) = \begin{cases} \frac{3}{2}t & , \quad 0 \leq t \leq 4, \\ \frac{3}{2}t^2 - 18t + 54 & , \quad 4 < t \leq 6, \end{cases}$$

where t denotes the time (in seconds) after the start of the first test and $f(t)$ denotes Terra's speed (in metres per second) at time t . The repetition of the test can be represented by the relationship given by $f(t) = f(t+6)$ for $t \geq 0$.

- (v) Sketch the graph of $y = f(t)$ for $0 \leq t \leq 14$. [2]
 (vi) The distance covered by Terra is given by the area under the graph of $y = f(t)$. Find the distance travelled by Terra from $t = 4$ to $t = 11$. [2]

Source of Question: PJC/Prelim/2018/01/Q9

- 11 (a) The product of the first three terms of a convergent geometric progression is 1000. If 6 is added to the second term and 7 is added to the third term, the three terms are now consecutive terms of an arithmetic progression. Find the first term and common ratio of the geometric progression. [4]
- (b) (i) A man takes a loan of \$ P for a house from a bank at the beginning of a month. The interest rate is 0.5 % per month so that at the start of every month, the amount of outstanding loan is increased by 0.5%. Equal instalment is paid to the bank at the end of every month. Find his monthly instalment if he would like to repay the loan in 20 years, leaving your answer in the form $0.005P \frac{k^n}{k^n - 1}$, where k and n are constants to be determined. [5]
- (ii) If the man takes a loan of \$1 000 000 for the house from another bank which charges interest at the start of every month, he will have to pay a monthly instalment of \$10 000 at the end of every month over 20 years to repay the loan. Find the monthly interest rate charged by this bank. [3]

Source of Question: EJC/Prelim/2018/01/Q8 modified

- 12 (a) It is given that $S_n = \frac{1}{2} - \frac{n}{(n+1)!}$.
- (i) Show that S_n converges to a limit L , where L is to be determined. [2]
- (ii) Find the least integer value of n such that S_n differs from L by less than 10^{-10} . [2]
- (b) (i) Suppose that f is a continuous, strictly decreasing function defined on $[1, \infty)$, with $f(x) > 0$, $x \geq 1$. According to the Maclaurin-Cauchy test, the infinite series $\sum_{n=1}^{\infty} f(n)$ is convergent if and only if the integral $\int_1^{\infty} f(x) dx$ is finite. By applying the Maclaurin-Cauchy test on the function f defined by $f(x) = \frac{1}{x}$, $x \geq 1$, determine if the infinite series $\sum_{n=1}^{\infty} \frac{1}{n}$ is convergent. [2]
- (ii) Let p be a positive number. By considering the Maclaurin-Cauchy test, show that if $p > 1$, the infinite series $1 + \frac{1}{2^p} + \frac{1}{3^p} + \dots + \frac{1}{n^p} + \dots$ is convergent. [2]

Source of Question: DHS/Prelim/2018/01/Q10

13 Duncan, an aspiring marathon runner, embarks on the following training regime:

1. On Day 1, he runs 5 km.
2. On Day 2, he runs $(1 + \alpha)$ times the distance ran in the previous day.
3. On Day 3, he runs α times the distance ran in the previous day.

You may take α to be a positive constant.

For each subsequent day, Duncan repeats parts 2 and 3 of the training regime, in that order. Thus on Day 4, he runs $(1 + \alpha)$ times the distance ran on Day 3; and on Day 5, he runs α times the distance ran on Day 4, and so on.

- (i) The distances that Duncan runs on odd-numbered days follows a geometric progression. State its common ratio in terms of α . [1]
- (ii) By considering just the distances that Duncan runs on even-numbered days, find the range of values of α such that the total distance ran on all even-numbered days is finite. [2]
- (iii) By considering the total distances ran on both odd and even numbered days, determine the theoretical maximum total distance, expressing your answer in terms of α . [2]

Duncan decides to fix $\alpha = 0.65$.

- (iv) (a) Show that the distance he ran on Day 10 is 10.915 km, correct to the nearest 0.001 km. [1]
- (b) The distance he ran on Day n first exceeds 42.195 km. Find the value of n . [3]
- (c) Duncan aims to complete his training regime on Day n , and instead of following the regime for that day, he plans to only run exactly 42.195 km on that day. Find the total distance that Duncan would have covered at the end of his training regime. [4]